

EXIT EXAM 2017

Software Engineering Exit Exam

Multiple Choice Questions (99 Questions)

1. What is the first step in managing a change to software requirements?

- A) Communicate the change to stakeholders
 - B) Assess the impact of the change
 - C) Establish a change management process
 - D) Implement the change immediately
-

2. In the context of debugging, what is the purpose of a breakpoint?

- A) To indicate the start of a loop
 - B) To pause the execution of a program
 - C) To mark the end of a function
 - D) To indicate an error in the code
-

3. Which of the following SQL statements is used to select all columns and records from a table named "Students"?

- A) SELECT ALL FROM Students;
 - B) SELECT Fields FROM Students;
 - C) SELECT * From Students;
 - D) SELECT Students;
-

4. Which one of the following Application Layer Protocol is used in Web Development is a Query-based API that allows clients to request only needed data?

- A) HTTP
 - B) GraphQL
 - C) HTTPS
 - D) REST
-

5. Which of the following OSI model layer handles issues such as network transparency and resource allocation?

- A) Transport
 - B) Network
 - C) Application
 - D) Data link
-

6. Which one of the following classes represents a top-level window with a title and border decorations?

- A) Container
 - B) Panel
 - C) Frame
 - D) Window
-

7. Which one of the following is the main purpose of conducting interviews in requirements elicitation?

- A) To gather a wide variety of user opinions through surveys
 - B) To gather in-depth insights from individual stakeholders
 - C) To develop detailed documentation of the system
 - D) To design and prototype the software system
-

8. What is the correct declaration of an integer array of 5 elements in C++?

- A) `int[] arr = {1, 2, 3, 4, 5};`
 - B) `int arr(5);`
 - C) `int arr[5] = 1, 2, 3, 4, 5;`
 - D) `int arr[5];`
-

9. Which one of the following Types of Agile testing Ensures new code changes does not break existing functionality?

- A) Regression Testing
 - B) Exploratory Testing
 - C) User Acceptance Testing (UAT)
 - D) Smoke Testing
-

10. Which of the following tools of software engineering enable tracking changes made to software configuration files during installation and configuration?

- A) Version control systems
 - B) Document generation tools
 - C) Package managers
 - D) Installation tools
-

11. Which risk response strategy involves transferring the impact of a risk to a third party, such as through insurance or outsourcing?

- A) Risk Mitigation
 - B) Risk Acceptance
 - C) Risk Transference
 - D) Risk Avoidance
-

12. Which one of the following types of operating systems kernel is a Minimalist kernel that delegates resource management to applications?

- A) Microkernel
 - B) Exokernel
 - C) Hybrid Kernel
 - D) Monolithic Kernel
-

13. Which one of the following Performance Optimization Techniques helps to Distribute content across global servers to reduce latency?

- A) Load Balancing
 - B) Content Delivery Network
 - C) Lazy Loading
 - D) Caching
-

14. Which Normal Form eliminates partial dependencies?

- A) Third Normal Form (3NF)
 - B) Second Normal Form (2NF)
 - C) First Normal Form (1NF)
 - D) Boyce-Codd Normal Form (BCNF)
-

15. What will be the output of the following code?

```
int day = 2;  
switch (day) {  
    case 1: cout << "Monday"; break;  
    case 2: cout << "Tuesday"; break;  
    case 3: cout << "Wednesday"; break;  
    default: cout << "Invalid day"; break;  
}
```

- A) Wednesday
 - B) Tuesday
 - C) Monday
 - D) Invalid day
-

16. Which architectural style is most suitable for a large-scale software system with independent, scalable components?

- A) Microservices Architecture
 - B) Layered Architecture
 - C) Client-Server Architecture
 - D) Monolithic Architecture
-

17. What should a programmer do when they encounter an error they do not understand?

- A) Panic and quit the project.
 - B) Assume they are not smart enough to fix it.
 - C) Search for solutions in documentation, forums, or ask for help.
 - D) Ignore the error and hope it goes away.
-

18. Among the listed Encryption Technology which one is used to secure database storage?

- A) SHA-256
 - B) Hashing Algorithm
 - C) AES
 - D) RSA
-

19. Imagine you want to design a database to store information about orders in online shopping. Which tables and relationships would be most appropriate?

- A) One table with all products details (number, name, type, customers, etc.)
 - B) Multiple tables with redundant data entries for faster searching
 - C) Separate tables for Products (number, name) and Customers (name), with relationships
 - D) Separate tables for Products (number), Customers (name), with no relationships
-

20. Which of the following function checks if the end of the file has been reached in C++ file handling?

- A) `getline()`
 - B) `eof()`
 - C) `close()`
 - D) `seekg()`
-

21. Which of the following is required by data privacy laws like GDPR and CCPA?

- A) Users must be informed about how their data is being used
 - B) Developers are allowed to collect data without user consent
 - C) User data must only be stored for 6 months
 - D) Users must provide anonymous data for app functionality
-

22. Which software engineering practice focuses on identifying and mitigating potential risks during the development process?

- A) Deployment planning
 - B) Risk management
 - C) Version control
 - D) User interface design
-

23. Ethiopian airline wants to optimize flight scheduling to minimize delays. Which algorithm design approach is BEST suited with this task?

- A) Sorting flight data by departure time.
 - B) Implementing a simple linear search for flight numbers.
 - C) Designing a graph-based algorithm to model flight connections and dependencies.
 - D) Randomly assigning flight departure times.
-

24. Which one of the following is a critical feature when designing for system scalability in a large-scale software system?

- A) Using a monolithic codebase for simplicity
 - B) Limiting API access to avoid traffic
 - C) Horizontal scaling through independent services
 - D) Using a single database for all services
-

25. Given a complex software development project, how would you design a Work Breakdown Structure (WBS) to ensure efficient task allocation?

- A) Focus only on the coding phase while ignoring testing and deployment.
 - B) Create a hierarchical breakdown of tasks based on project deliverables.
 - C) Assign all tasks to a single team to ensure consistency.
 - D) Divide the work into as many random tasks as possible.
-

26. Which of the following is the primary security concern in choosing a web-based solution for a business?

- A) The server of the websites.
 - B) The choice of JavaScript framework.
 - C) The number of images used on the website
 - D) Protection against cyber threats like SQL Injection and XSS.
-

27. Imagine you have been assigned to schedule a software development project with multiple dependencies. What approach would you take to create an effective timeline?

- A) Use a Gantt chart and critical path method to plan dependencies
 - B) Ignore dependencies and focus only on coding speed
 - C) Allow developers to choose their own timelines without oversight
 - D) Assign equal deadlines to all tasks regardless of complexity
-

28. Which one of the following deadlock condition occurs when a resource can only be used by one process at a time?

- A) Circular Wait
 - B) Hold and Wait
 - C) Mutual Exclusion
 - D) No Preemption
-

29. Which one of the following Malware Type Encrypts user files and demands payment?

- A) Viruses
 - B) Ransomware
 - C) Rootkits
 - D) Worms
-

30. Which one of the following types of Machine Learning Learns from labeled data?

- A) Semi-Supervised Learning
 - B) Unsupervised Learning
 - C) Reinforcement Learning
 - D) Supervised Learning
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31. Which one of the following levels of Software Testing test interactions between modules?

- A) System Testing
 - B) Unit Testing
 - C) Integration Testing
 - D) User Acceptance Testing
-

32. Suppose you are designing a data center network that requires minimal latency and high redundancy. Which topology would best meet the requirement?

- A) Bus
 - B) Mesh
 - C) Star
 - D) Hybrid
-

33. Which one of the following Static software Testing type is a formal and well-defined type of static test which was first introduced in 1972 in IBM by Michael Fagan?

- A) Inspection
 - B) Technical review
 - C) Walk-through
 - D) Informal review
-

34. Which one of the following types of machine learning algorithms is best suited for demand forecasting and Predictive maintenance problems?

- A) Time-Series Forecasting
 - B) Regression
 - C) Classification
 - D) Clustering
-

35. Which method is used in JavaScript to change the content of an HTML element dynamically?

- A) document.write()
 - B) window.alert()
 - C) document.getElementById().innerHTML
 - D) console.log()
-

36. Which one of the following statement is TRUE?

- A) An overriding method can declare that it throws checked exceptions that are not thrown by the method it is overriding.
 - B) Private methods cannot be overridden in subclasses.
 - C) The parameter list of an overriding method can be a subset of the parameter list of the method that it is overriding.
 - D) A subclass can override any method in a superclass.
-

37. Which of the following CSS style properties are inherited, meaning they are transmitted from parent to child elements?

- A) Margins style
 - B) Height style
 - C) Borders style
 - D) Colors style
-

38. What is the primary method used to estimate the maintenance effort for software systems?

- A) Lines of Code (LOC)
 - B) Number of security patches required
 - C) Function Points (FP)
 - D) Annual Maintenance Cost (AMC)
-

39. What is the purpose of a foreign key in database design?

- A) To improve indexing performance.
 - B) To enforce referential integrity between related tables.
 - C) To store duplicate records.
 - D) To uniquely identify a record within a table.
-

40. Which of the following layer in the TCP/IP model provides a logical connection between a source host and a destination host?

- A) Data link
 - B) Network
 - C) Transport
 - D) Application
-

41. Which one of the following algorithm would you use to check if a number is even or odd?

- A) If-else statement
 - B) Merge sort
 - C) Sorting algorithm
 - D) Binary search
-

42. Which of the following is a type of AI agent that maintains an internal model of the world, which is updated by percepts as they are received?

- A) Simple reflex agent
 - B) Learning agent
 - C) Model-based agent
 - D) Utility-based agent
-

43. A software company wants to implement a new data security policy for newly developed system. The most effective way to ensure employee awareness and compliance is to:

- A) Rely on employees to read and understand the policy on their own time.
 - B) Delegate responsibility for policy enforcement to IT security personnel.
 - C) Provide comprehensive training on the policy and its importance.
 - D) Distribute the policy document via email and require a signature for receipt.
-

44. Which sorting algorithm has the worst-case time complexity of $O(n^2)$?

- A) Merge Sort
 - B) Bubble Sort
 - C) Quick Sort
 - D) Heap Sort
-

45. Why learning Artificial Intelligence is so important in today's world?

- A) Learning AI once is enough to know all future developments.
 - B) AI technologies have no impact on human beings.
 - C) AI is a rapidly evolving field, requiring continuous learning to stay updated.
 - D) AI has limited applications, so further learning is necessary.
-

46. Which one of the following is used for State management in web applications development?

- A) React Router
 - B) Redux
 - C) Axios
 - D) Content Delivery Network
-

47. Which of the following is the correct way to declare a function that takes two integers and returns their sum?

- A) `sum(int a, int b) { return a + b; }`
 - B) `int sum(a, b) { return a + b; }`
 - C) `sum(int a, b) { return a + b; }`
 - D) `int sum(int a, int b) { return a + b; }`
-

48. Which one of the following deadlock condition occurs when a process holding at least one resource is waiting to acquire additional resources held by other processes?

- A) Hold and Wait
 - B) No Preemption
 - C) Mutual Exclusion
 - D) Circular Wait
-

49. Among the listed component of Android Application which one helps to declare permissions such as accessing sensitive data (GPS, Camera, and Storage)?

- A) Views and ViewGroups
 - B) Content Provider
 - C) Intents
 - D) AndroidManifest.xml
-

50. Which one of the following is a code smell in software?

- A) Frequent code reviews
 - B) Consistent naming conventions
 - C) High code test coverage
 - D) Long method length
-

51. Which of the following topology, the central server will provide the connectivity for all pair of nodes willing to communicate with each other?

- A) Bus
 - B) Mesh
 - C) Star
 - D) Ring
-

52. Write a program to find the sum of the first n natural numbers (1 to n). What is the correct formula to calculate this sum?

- A) $n * (n + 1) / 2$
 - B) $n * (n + 2) / 2$
 - C) $n * (n - 1) / 2$
 - D) $n + (n - 1) / 2$
-

53. Which of the following is not user defined data type?

- A) Structure
 - B) Union
 - C) Enumeration
 - D) Integer
-

54. Which of the following is correct regarding the difference between breadth first search and depth first search?

- A) Breadth first search requires less memory compared to depth first search.
 - B) Breadth first search is slower than depth first search
 - C) Breadth first search is more faster than depth first search
 - D) Depth first search is useful in finding shortest path, while breadth first search is not.
-

55. What is the primary focus of refactoring in software development?

- A) Rewriting the entire software to improve its performance
 - B) Enhancing the external user interface
 - C) Adding new functionalities
 - D) Improving internal code structure without changing external behavior
-

56. Which one of the following process Synchronization Mechanism Ensures only one process accesses a shared resource at a time?

- A) Mutex (Mutual Exclusion Locks)
 - B) Semaphores
 - C) Read-Write Locks
 - D) Monitors and Condition Variables
-

57. Which of the following models is NOT considered as the state-of-the-art software process model?

- A) Extreme Programming (XP)
 - B) Waterfall model
 - C) Agile Model
 - D) Spiral model
-

58. Which of the following project management tool is used for scope management?

- A) Critical path analysis
 - B) Gantt Chart
 - C) Net present value
 - D) Work breakdown structures
-

59. Which of the following activity is not conducted in software planning?

- A) Organizational and Resource Planning
 - B) Managing Stage Boundaries
 - C) Cost Estimation and Budgeting
 - D) Project Plan Development and Execution
-

60. Which one of the following object oriented feature restricts direct access to certain details, ensuring data security and integrity?

- A) Polymorphism
 - B) Encapsulation
 - C) Inheritance
 - D) Abstraction
-

61. Which one of the following is the purpose of AJAX in web development?

- A) To handle server-side database interactions
 - B) To create static HTML pages
 - C) To make asynchronous requests to the server and update the page dynamically
 - D) To apply styles to HTML elements
-

62. Which one of the following incident requires immediate action (e.g., system failure)?

- A) High
 - B) Low
 - C) Medium
 - D) Critical
-

63. Which one of the following types of firewall Tracks active connections and allows only legitimate traffic?

- A) Proxy Firewall
 - B) Cloud-Based Firewall
 - C) Stateful Inspection Firewall
 - D) Packet Filtering Firewall
-

64. What is the primary role of CSS in web development?

- A) To structure the webpage content
 - B) To style and layout the webpage elements
 - C) To add interactivity to the page
 - D) To interact with the server
-

65. What is the first step in developing a medium-scale application that solves real world problems?

- A) Writing code immediately
 - B) Choosing a programming language randomly
 - C) Copying code from online sources
 - D) Analyzing requirements and designing system architecture
-

66. Which one of the following Vulnerability Injects JavaScript to steal user data?

- A) Insecure Deserialization
 - B) Cross-Site Scripting (XSS)
 - C) Spyware
 - D) Buffer Overflow
-

67. Suppose you are developing an application that involves real-time data analysis and requires efficient insertion and deletion of data. Which of the following data structures would be best suited for this application?

- A) Stack
 - B) Queue
 - C) Binary tree
 - D) Hash table
-

68. Which one of the following Process Scheduling Algorithms Executes processes in order of arrival?

- A) Round Robin (RR)
 - B) First-Come, First-Served (FCFS)
 - C) Priority Scheduling
 - D) Shortest Job Next (SJN)
-

69. Which of the following Software Architectural Styles is typically used in systems where high responsiveness and real-time processing are needed?

- A) Monolithic Architecture
 - B) Microservices Architecture
 - C) Event-Driven Architecture
 - D) Serverless Architecture
-

70. Which of the following is not the function of OSI model presentation layer?

- A) Encryption
 - B) Compression
 - C) Translation
 - D) Synchronization
-

71. Which one of the following is a Security Concern in Mobile Networks where attackers steal session cookies?

- A) Man-in-the-Middle (MitM) attacks
 - B) SQL Injection
 - C) Session hijacking
 - D) Ransomware
-

72. Which one of the following is a type of software defect?

- A) Repeated code segments in the program
 - B) Null reference error in a function
 - C) An unhandled exception in the code
 - D) Performance degradation due to excessive memory usage
-

73. Which one of the following Core Objectives of Cryptography Verifies the identity of communicating parties?

- A) Integrity
 - B) Authentication
 - C) Confidentiality
 - D) Non-repudiation
-

74. Which of the following design pattern is used when the algorithm for creating a complex object should be independent of the parts that make up the object and how they are assembled?

- A) Singleton pattern
 - B) Builder pattern
 - C) Factory Method pattern
 - D) Prototype pattern
-

75. Suppose you have entity, Samuel. Samuel is a player, his age is 25, his height is 1.98, his club is Manchester, and he lives in England, Manchester. Which of the following knowledge representation is best to represent it?

- A) Logical representation
 - B) Semantic network Representation
 - C) Rule based representation
 - D) Frame representation
-

76. What is the primary purpose of an Integrated Development Environment (IDE)?

- A) To write and debug code
 - B) To run mobile applications
 - C) To store user data
 - D) To manage hardware resources
-

77. Software engineering principles like KISS (Keep It Simple, Stupid) focus on achieving which one of the following?

- A) Faster development speed at the cost of quality
 - B) Improved code maintainability and reusability
 - C) Increased software complexity
 - D) More visually appealing user interfaces
-

78. Which of the following sorting algorithms follows the DIVIDE and CONQUER approach?

- A) Merge Sort
 - B) Bubble Sort
 - C) Selection Sort
 - D) Insertion Sort
-

79. Which one of the following Application Layer Protocol used in Web Development Enables real-time, two-way communication between client and server?

- A) REST
 - B) WebSocket
 - C) GraphQL
 - D) HTTP
-

80. Which one of the following types of Machine Learning Discovers patterns from unlabeled data?

- A) Unsupervised Learning
 - B) Reinforcement Learning
 - C) Supervised Learning
 - D) Semi-Supervised Learning
-

81. Which metric measures the time between two consecutive failures in a system?

- A) Defect Density
 - B) Mean Time to Repair
 - C) Test Coverage
 - D) Mean Time Between Failures
-

82. A software developer has completed coding a new feature for a project. Before integrating it with the main codebase, what is the MOST important step to ensure quality?

- A) Reviewing the code for potential errors and adherence to coding standards.
 - B) Writing extensive user documentation for the new feature.
 - C) Deploying the code directly to the production environment for testing.
 - D) Updating marketing materials to showcase the new feature.
-

83. Which one of the following types of machine learning algorithms is best suited for Email spam detection problem?

- A) Clustering
 - B) Time-Series Forecasting
 - C) Regression
 - D) Classification
-

84. Which one of the following is a key characteristic of a well-written functional requirement?

- A) It should focus only on performance issues
 - B) It should be vague to allow flexibility during development
 - C) It should be testable and measurable
 - D) It should describe the system architecture
-

85. Non-functional requirements typically focus on which of the following?

- A) Specific programming languages and frameworks to be used
 - B) Performance, security, usability, and scalability of the system
 - C) System behaviors like authentication and user interactions
 - D) Technical specifications for hardware
-

86. What method can be used by two computers to ensure that packets are not dropped because too much data is being sent too quickly?

- A) Encapsulation
 - B) Access method
 - C) Flow control
 - D) Response timeout
-

87. Which one of the following type of maintenance is used to refer to maintenance for fault repair?

- A) Adaptive maintenance
 - B) Function Points (FP)
 - C) Corrective maintenance
 - D) Perfective maintenance
-

88. Which loop is guaranteed to execute at least once, even if the condition is false?

- A) while loop
 - B) do-while loop
 - C) For each loop
 - D) for loop
-

89. How do you add new department column to Students table?

- A) MODIFY TABLE Students ADD department VARCHAR(255);
 - B) INSERT INTO Students ADD COLUMN department VARCHAR(255);
 - C) ALTER TABLE Students ADD COLUMN department VARCHAR(255);
 - D) UPDATE TABLE Students ADD COLUMN department VARCHAR(255);
-

90. Which one of the following Object-Oriented Software Design Principle refers that software entity should be open for extension but closed for modification?

- A) Interface Segregation Principle (ISP)
 - B) Single Responsibility Principle (SRP)
 - C) Liskov Substitution Principle (LSP)
 - D) Open/Closed Principle (OCP)
-

91. What is the primary goal of requirement engineering in software development?

- A) To write the code
 - B) To create the user interface
 - C) To define and manage stakeholder requirements
 - D) To design the software architecture
-

92. What is the main purpose of debugging?

- A) To remove all comments in the code.
 - B) To add more features without testing.
 - C) To make the code more complex.
 - D) To identify and fix errors in the code.
-

93. Which component of Android Application refers to a reusable UI component within an activity?

- A) Broadcast Receivers
 - B) Fragments
 - C) Activities
 - D) Content Providers
-

94. Which one of the following Authentication Method uses cryptographic certificates for secure authentication?

- A) Multi-Factor Authentication (MFA)
 - B) OTP
 - C) Public Key Infrastructure (PKI)
 - D) Biometric Authentication
-

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